

Lecture 5

Empirical Running Time



Announcements

- ❖ **Lab 1** on next Monday
 - Empirically benchmarking algorithms
- ❖ **Office hours** are up on course website
 - T: 12pm-4pm; F: 11am-1pm
 - Schedule an appointment individually or as a group
- ❖ **Project 1** problems released tomorrow
 - Choose your team of 3
 - Details up now on course website
 - Planning & Analysis writeup due next Friday 9/18 @ 11:59PM

Today

- ❖ Running time
- ❖ Empirically measuring running time
- ❖ Benchmarking algorithms

Running Time

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- ❖ Questions to consider:
 - What does "long" mean? Wall-clock time?
 - How long to run on which input? Several of them? How do we choose?
 - Does running time vary by properties of the input (particularly the size)?
 - What if my machine is slower than another machine?
 - How do I guarantee a fair comparison between two algorithms?

Measuring Running Time

Two strategies:

- ❖ **Empirical**

- For a fixed compute environment, run an implementation of an algorithm on several inputs of varying properties and record execution time

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Jupyter

- ❖ Let's move over to [a notebook](#)

Next Time

- ❖ Asymptotic running time
- ❖ DSP 5.4-5.10